1. Write a program to check whether a given number is present in array or not (Linear search).

```c
#include <stdio.h>
int main (){
    int size;
    printf("enter the size of array : ");
    scanf("%d",&size);
    int arr[size];
    for (int i=0;i<size;i++){
        scanf("%d",&arr[i]);
    }
    int target;
    printf("please enter the the number to be checked ");
    scanf("%d",&target);
    int flag=0;
    int i=0;
    for (i=0;i<size;i++){
        if (arr[i]==target){
            flag=1;
            break;
        }
    }
    if (flag==1){printf("the number is at index : %d\n",i);}
    else printf("not found\n");
return 0;
}
```

```
~/Documents/dslab  nvim assignment2_Q1.c
~/Documents/dslab  g++ assignment2_Q1.c
~/Documents/dslab  ./a.out
enter the size of array : 6
1 2 3 4 5 6
please enter the the number to be checked 1
the number is at index : 0
~/Documents/dslab  ./a.out
enter the size of array : 5
1 2 3 4 6
please enter the the number to be checked 5
not found
~/Documents/dslab  ./a.out
enter the size of array : 4
1 8 63
1
please enter the the number to be checked 6
not found
```

2. Write a program to get second maximum and second minimum elements in an array.

```c
#include <stdio.h>
#include <limits.h>

int main() {
    int size;
    printf("Enter the size of the array: ");
    scanf("%d", &size);

    if (size < 2) {
        printf("Array must have at least two elements.\n");
        return 1;
    }

    int arr[size];
    printf("Enter the elements of the array:\n");
    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }

    int max = INT_MIN;
    int secondMax = INT_MIN;
    int min = INT_MAX;
    int secondMin = INT_MAX;

    for (int i = 0; i < size; i++) {
        if (arr[i] > max) {
            secondMax = max; // The old max is now the second max
            max = arr[i];    // We have a new max
        } else if (arr[i] > secondMax && arr[i] != max) {
            secondMax = arr[i];
        }

        if (arr[i] < min) {
            secondMin = min; // The old min is now the second min
            min = arr[i];    // We have a new min
        } else if (arr[i] < secondMin && arr[i] != min) {
            secondMin = arr[i];
        }
    }

    if (secondMax == INT_MIN) {
        printf("No second maximum found (all elements might be the same).\n");
    } else {
        printf("Second maximum = %d\n", secondMax);
    }

    if (secondMin == INT_MAX) {
        printf("No second minimum found (all elements might be the same).\n");
    } else {
        printf("Second minimum = %d\n", secondMin);
    }

    return 0;
}
```

```
~/Documents/dslab  nvim assignment2_Q2.c
~/Documents/dslab  g++ assignment2_Q2.c
~/Documents/dslab  ./a.out
enter the size of array : 2
1 2
no second maximum found
no second minimum found
~/Documents/dslab  ./a.out
enter the size of array : 5
5 5 4 3 2
no second maximum found
second minimum = 3
~/Documents/dslab  nvim assignment2_Q2.c
~/Documents/dslab  g++ assignment2_Q2.c
~/Documents/dslab  ./a.out
Enter the size of the array: 5
Enter the elements of the array:
1 2 3 4 5
Second maximum = 4
Second minimum = 2
~/Documents/dslab  ./a.out
Enter the size of the array: 3
Enter the elements of the array:
 2 2
1
Second maximum = 1
Second minimum = 2
~/Documents/dslab  ./a.out
Enter the size of the array: 2
Enter the elements of the array:
1 2
Second maximum = 1
Second minimum = 2
~/Documents/dslab  ./a.out
Enter the size of the array: 6
Enter the elements of the array:
1 2 3
4 5 6
Second maximum = 5
Second minimum = 2
~/Documents/dslab
```

3. Write a program to perform insertion (any location), deletion (any location) and traversal in an array.

```c
#include <stdio.h>
int main (){
    int size;
    printf("enter the size of array : ");
    scanf("%d",&size);
    int arr[100];
    for (int i=0;i<size;i++){
        scanf("%d",&arr[i]);
    }
    printf("array is : ");
    for (int i=0;i<size;i++){
        printf("%d ",arr[i]);
    }
    printf("\n");
    int pos, num;
    printf("enter position (0 to %d) and number for insertion: ", size);
    scanf("%d %d", &pos, &num);
    for (int i=size;i>pos;i--){
        arr[i] = arr[i-1];
    }
    arr[pos] = num;
    size++;
    printf("after insertion: ");
    for (int i=0;i<size;i++){
        printf("%d ",arr[i]);
    }
    printf("\n");
    printf("enter position (0 to %d) for deletion: ", size-1);
    scanf("%d",&pos);
    for (int i=pos;i<size-1;i++){
        arr[i] = arr[i+1];
    }
    size--;
    printf("after deletion: ");
    for (int i=0;i<size;i++){
        printf("%d ",arr[i]);
    }
    printf("\n");
    return 0;
}
```

```
second minimum = 2
~/Documents/dslab  nvim assignment2_Q3.c
~/Documents/dslab  g++ assignment2_Q3.c
~/Documents/dslab  ./a.out
enter the size of array : 5
1 2 3 4 5
array is : 1 2 3 4 5
enter position (0 to 5) and number for insertion: 2
5
after insertion: 1 2 5 3 4 5
enter position (0 to 5) for deletion: 5
after deletion: 1 2 5 3 4
~/Documents/dslab  ./a.out
enter the size of array : 4
1 8 9 7
array is : 1 8 9 7
enter position (0 to 4) and number for insertion: 4
444
after insertion: 1 8 9 7 444
enter position (0 to 4) for deletion: 1
after deletion: 1 9 7 444
~/Documents/dslab
```

4. Write a menu driven program to perform addition, multiplication and substraction of 2 arrays.

```c
#include <stdio.h>
int main (){
    int n;
    printf("enter size of arrays : ");
    scanf("%d", &n);
    int arr1[n], arr2[n];
    printf("enter elements of array 1:\n");
    for (int i=0;i<n;i++) scanf("%d",&arr1[i]);
    printf("enter elements of array 2:\n");
    for (int i=0;i<n;i++) scanf("%d",&arr2[i]);
    int choice;
    printf("1: Addition\n2: Subtraction\n3: Multiplication\nenter your choice: ");
    scanf("%d", &choice);
    printf("result: ");
    for (int i=0; i<n; i++) {
        if (choice==1) printf("%d ", arr1[i]+arr2[i]);
        else if (choice==2) printf("%d ", arr1[i]-arr2[i]);
        else if (choice==3) printf("%d ", arr1[i]*arr2[i]);
        else printf("invalid choice");
    }
    printf("\n");
    return 0;
}
```

```
~/Documents/dslab  nvim assignment2_Q4.c
~/Documents/dslab  g++ assignment2_Q4.c
~/Documents/dslab  ./a.out
enter size of arrays : 4
enter elements of array 1:
1
2
3
4
enter elements of array 2:
2
3
4
5
1: Addition
2: Subtraction
3: Multiplication
enter your choice: 1
result: 3 5 7 9
~/Documents/dslab  ./a.out
enter size of arrays : 1 2 3 4
enter elements of array 1:
enter elements of array 2:
1: Addition
2: Subtraction
3: Multiplication
enter your choice: result: invalid choice
~/Documents/dslab  ./a.out
enter size of arrays : 5
enter elements of array 1:
5 5 6 9 8
enter elements of array 2:
1 2 3 8 7
1: Addition
2: Subtraction
3: Multiplication
enter your choice: 2
result: 4 3 3 1 1
~/Documents/dslab
```

5. Write a program to perform sorting while merging (Merge two sorted arrays into one sorted array).

```c
#include <stdio.h>
int main (){
    int n1, n2;
    printf("enter size of first array : ");
    scanf("%d",&n1);
    printf("enter size of second array : ");
    scanf("%d",&n2);
    int arr1[n1], arr2[n2], merged[n1+n2];
    printf("enter sorted elements of array 1:\n");
    for (int i=0;i<n1;i++) scanf("%d",&arr1[i]);
    printf("enter sorted elements of array 2:\n");
    for (int i=0;i<n2;i++) scanf("%d",&arr2[i]);
    int i=0,j=0,k=0;
    while(i<n1 && j<n2){
        if(arr1[i]<=arr2[j]) merged[k++] = arr1[i++];
        else merged[k++] = arr2[j++];
    }
    while(i<n1) merged[k++] = arr1[i++];
    while(j<n2) merged[k++] = arr2[j++];
    printf("merged sorted array: ");
    for (int i=0;i<n1+n2;i++) printf("%d ",merged[i]);
    printf("\n");
    return 0;
}
```

```
result: 4 5 5 1 1
~/Documents/dslab  g++ assignment2_Q5.c
~/Documents/dslab  ./a.out
enter size of first array : 5
enter size of second array : 1
enter sorted elements of array 1:
2
3
4
5
6
enter sorted elements of array 2:
5
merged sorted array: 2 3 4 5 5 6
~/Documents/dslab  ./a.out
enter size of first array : 5
enter size of second array : 5
enter sorted elements of array 1:
1 3 5 7 8
enter sorted elements of array 2:
2 4 6 9 11
merged sorted array: 1 2 3 4 5 6 7 8 9 11
~/Documents/dslab  ./a.out
enter size of first array : 4
enter size of second array : 4
enter sorted elements of array 1:
1 2 3 4
enter sorted elements of array 2:
5 6 7 8
merged sorted array: 1 2 3 4 5 6 7 8
~/Documents/dslab  nvim assignment2_Q4.c
```

6. Write the above programs (1,2, and 3) using functions and call by address only.

```c
#include <stdio.h>

void linear_search(int *arr, int size, int target) {
    int flag=0;
    int i=0;
    for (i=0;i<size;i++){
        if (arr[i]==target){
            flag=1;
            break;
        }
    }
    if (flag==1){printf("the number is at index : %d\n",i);}
    else printf("not found\n");
}

void second_max_min(int *arr, int size){
    int max, secondMax, min, secondMin;
    max = secondMax = arr[0];
    min = secondMin = arr[0];
    for (int i=1;i<size;i++){
        if (arr[i] > max) max = arr[i];
        if (arr[i] < min) min = arr[i];
    }
    int foundMax = 0, foundMin = 0;
    for (int i=0;i<size;i++){
        if (arr[i] > secondMax && arr[i] != max){
            secondMax = arr[i];
            foundMax = 1;
        }
        if (arr[i] < secondMin && arr[i] != min){
            secondMin = arr[i];
            foundMin = 1;
        }
    }
    if (foundMax) printf("second maximum = %d\n", secondMax);
    else printf("no second maximum found\n");
    if (foundMin) printf("second minimum = %d\n", secondMin);
    else printf("no second minimum found\n");
}
```

```c

void array_ops(int *arr, int *size) {
    printf("array is : ");
    for (int i=0;i<*size;i++){
        printf("%d ",arr[i]);
    }
    printf("\n");
    int pos, num;
    printf("enter position (0 to %d) and number for insertion: ", *size);
    scanf("%d %d", &pos, &num);
    for (int i=*size;i>pos;i--){
        arr[i] = arr[i-1];
    }
    arr[pos] = num;
    (*size)++;
    printf("after insertion: ");
    for (int i=0;i<*size;i++){
        printf("%d ",arr[i]);
    }
    printf("\n");
    printf("enter position (0 to %d) for deletion: ", *size-1);
    scanf("%d",&pos);
    for (int i=pos;i<*size-1;i++){
        arr[i] = arr[i+1];
    }
    (*size)--;
    printf("after deletion: ");
    for (int i=0;i<*size;i++){
        printf("%d ",arr[i]);
    }
    printf("\n");
}

int main() {
    int size;
    printf("enter the size of array : ");
    scanf("%d",&size);
    int arr[100];
    for (int i=0;i<size;i++){
        scanf("%d",&arr[i]);
    }
    int target;
    printf("please enter the the number to be checked ");
    scanf("%d",&target);
    linear_search(arr, size, target);
    second_max_min(arr, size);
    array_ops(arr, &size);
    return 0;
```

NORMAL assignment2_06.c

```
~/Documents/dslab  g++ assignment2.core
~/Documents/dslab  ./a.out
enter the size of array : 6
1 2 3 4 5 6
please enter the the number to be checked 5
the number is at index : 4
second maximum = 5
no second minimum found
array is : 1 2 3 4 5 6
enter position (0 to 6) and number for insertion: 0
2
after insertion: 2 1 2 3 4 5 6
enter position (0 to 6) for deletion: 3
after deletion: 2 1 2 4 5 6
~/Documents/dslab  ./a.out
enter the size of array : 5
2 8 4 9 34
please enter the the number to be checked 5
not found
second maximum = 9
no second minimum found
array is : 2 8 4 9 34
enter position (0 to 5) and number for insertion: 2
55
after insertion: 2 8 55 4 9 34
enter position (0 to 5) for deletion: 6
after deletion: 2 8 55 4 9
```